Silicone Adhesives – General Data

1. General

WARNING THIS PROCEDURE USES MATERIALS THAT MAY BE DANGEROUS TO THE PERSONNEL'S HEALTH AND/OR THE ENVIRONMENT. SUFFICIENT PRECAUTIONS MUST BE TAKEN AT ANY STEP OF USE. IF NOT, THIS CAN CAUSE DAMAGE TO THE ENVIRONMENT AND DANGEROUS EFFECTS ON THE PERSONNEL'S HEALTH AND SAFETY.

This standard specifies the procedure and requirements for bonding aircraft parts with PSA 529/SRC–18 two part adhesive.

A. Materials and Equipment

- NAPHTHA
- Suitable grooved work holder
- Disposable wax-free paperboard containers
- Eye Dropper
- Clean cotton gloves
- Masking tape
- Weighing scale; capable of weighing to within + or – 0.1 grams
- PSA 529/SRC–18
- Self-adhesive labels (refer to Figure 1)
- Disposable wooden tongue depressor

B. Surface Treatment

1) Aluminum alloy parts must have bonding surfaces primed with fluid resistant epoxy primer.

2) Bonding surfaces of cadmium plated parts must be primed with fluid resistant epoxy primer.

3) Precautions must be taken and masking employed where necessary to ensure that plexiglass parts have only their bonding surfaces scuffed.

C. Safety Precautions

1) Adhesive, primer and solvents must be kept away from fire and other sources of ignition.

2) Sufficient ventilation must be provided when using solvent, primer or adhesive in confined areas.

3) Skin contact with solvent, primer and adhesive must be avoided; wear protective gloves.

4) Protective hand cream must not be used as it may cause contamination of cleaned or adhesive coated surfaces.

D. Cleaning the Bonding Surfaces

1) Surfaces to be bonded must be clean and completely free of contamination such as dirt, mould release agent, grease or finger marks.

2) When cleaning bonding surfaces with solvent saturate a clean lint-free cloth with solvent, as specified, and thoroughly scrub the bond surfaces. Wipe dry with a clean lint-free cloth before the solvent evaporates.

3) Keep solvent cloth saturated and turn frequently to avoid reusing a dirty area.

4) When cleaning or applying adhesive to long lengths of silicone rubber extrusion, a suitable grooved work holder should be used to hold and support part.

5) Cleaned, primed or adhesive coated surfaces must not be touched with bare hands or otherwise subjected to contamination. Wear clean cotton gloves when handling prepared surfaces.

6) Refer to for cleaning and preparation of various materials.

E. Preparation of Adhesives

Note PSA 529/SRC–18 adhesive is received in kits from the supplier. Each kit includes resin in a 4 fl. oz. glass bottle and a 3.3 ml glass bottle of catalyst.

1) Remove cap from the bottle of adhesive resin.

2) Add the contents of the bottle of catalyst directly into the bottle of resin. The kit supplies the resin and catalyst in the correct mixing ratio.

3) Using the disposable wooden tongue depressor, stir the resin/catalyst mixture thoroughly to obtain a homogeneous air-free mix. Immediately cap and close the bottle tightly.

4) Label the bottle according to Figure 1 and stamp with the expiry date (mixing date + 3 months pot life).

Shelf Life Adhesive Label
Figure 1

Table 1 Solvent Selection For General Materials

MATERIAL TO BE CLEANED	SOLVENT (Note 1)
Electrical connections	IPA
Bare metal	DS108, Diestone DLS, Diestone HFP or MEK
Primed or painted metal parts	DS108, Diestone DLS, Diestone HFP or MEK (Note 1)
Unprimed plastic parts (Note 2)	NAPHTHA
Unprimed fiberglass or phenolic (Note 3)	DS108 or MEK
Thermoplastic (Note 4)	Methanol or Naphtha
Thermoset plastic (Note 5)	MEK
Rulon A	DS108
Rubber (Note 6) except silicone	Naphtha

Table 1 Solvent Selection For General Materials (continued)

MATERIAL TO BE CLEANED	SOLVENT (Note 1)
Silicone rubber	IPA
Porous materials (Note 7)	Do not clean porous materials in any way. If the surfaces are contaminated, do not use the materials for bonding.

Note 1 – Use DS108 for cleaning aircraft exterior before painting.

Note 2 – Plastic parts include Formica, Arborite, etc. Plastic parts do not include Kevlar and fiberglass.

Note 3 – Fiberglass includes Kevlar laminates and composites. Phenolic laminates do not include Formica, Arborite, etc.

Note 4 – Thermoplastic: acrylic (Plexiglas), Acrylic/PVC (kydex), nylon (Zytel), polycarbonate (Lexan), PTFE (teflon), PVC/PVF (Tedlar).

Note 5 – Thermoset plastics: fibre-reinforced epoxy, polyester or phenolic composites.

Note 6 – Rubber: neoprene, Buna-N (nitrile, NBR).

Note 7 – Porous materials: Velcro, fabrics, balsa, cork, etc.

2. Bonding

A. Application

1) Apply masking tape adjacent to area to be bonded to prevent adhesive contamination of the adjacent surfaces.

2) Ensure areas to be bonded are clean (refer to Para 1.D and Table 1 — Solvent Selection For General Materials).

3) Pour required amount of catalyzed adhesive into suitable wax-free container. Immediately re-cap bottle.

4) Apply a thin uniform coat (0.010 to 0.015 inch) to both joining surfaces using a bristle brush of suitable size. Ensure that both surfaces are completely covered, including butt ends of silicone foam extrusions.

Note A grooved holder may be used to support the work (if applicable).

5) Allow adhesive to dry until a firm tack develops.

6) Join parts together and roll down with a rubber roller or stitch roller, or press firmly with fingers to ensure complete contact throughout.

Note Parts must be carefully positioned initially as bond must not be broken and repositioned.

7) Where the silicone extrusion tends to lift due to distortion, as around corners, apply moderate pressure by means of masking tape or other suitable means, until bond has cured sufficiently.

8) The bond must be allowed to cure for a minimum of 24 hours at 72°F (22°C) (room temperature) before further working the assembly or installing it in the aircraft.

Note Discard unused adhesive. Do not pour back into catalyzed bottle.

a) Bonded parts must have complete contact over the full bonding area.

b) Visual indication of poor adhesion is cause for rejection.

3. Storage

A. Store adhesive components at 60°F to 80°F (15°C to 26°C) with the storage precautions necessary for flammable materials.

B. Ensure that the containers are clearly stamped with the storage life expiry date. Refer to the MSDS and/or manufacturer's recommendations for the storage life of the adhesive components.

C. Keep containers of adhesives tightly closed when not in use.